

Luca Lanzillotta

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Education

California State Polytechnic University, Pomona

B.S. Electrical Engineering, Minor in Mathematics | GPA: 3.40/4.00

CA

Fall 2022 – Expected Spring 2027

Experience

F´ Flight Software Engineering Intern

June 2026 – August 2026

Jet Propulsion Laboratory (NASA/JPL)

Pasadena, CA

- Designed, implemented, and unit-tested F´ flight software components for spacecraft flight system architecture in C++ and Python, using CMake, Bash, and FPP.
- Performed hardware implementation and integration testing to validate flight software behavior on target systems.
- Collaborated with the flight software team on component interface reviews, code walkthroughs, and design reviews following JPL software assurance processes.
- Applied systems engineering principles across the software lifecycle, from requirements and component modeling through verification and validation.

Embedded Systems Engineer – SCALES

January 2025 – Present

Bronco Space Lab

Pomona, CA

- Designed, tested, and implemented a fault-tolerant power-supply PCB and a custom Linux-based ARM SOM carrier board for a mission-specific satellite subsystem, including high-speed routing and skew tuning for timing-sensitive interfaces.
- Created an integrated EPS + i.MX8X compute board variant, refining schematic connections, mounting-hole corrections, and production (Gerber/BOM/STEP) release outputs.
- Built the meta-scales-leviathan Yocto BSP for a PHYTEC i.MX8QXP SOM, modifying the Linux kernel, device tree, and U-Boot configuration to separate the Linux console from a dedicated F´ GDS communications UART.
- Migrated the fprime-scales-ref flight-software deployment from F´ 3.6.3 to F´ 4.2.2, resolving CMake, FPP, submodule, and Python-environment compatibility issues across i.MX8X and Jetson targets.
- Integrated F´ data products, topology, and downlink handling, and validated designs via simulation in Cadence OrCAD X, LTspice, and ngspice prior to hardware bring-up.

Hardware Design Lead & Project Manager – BILLEE

August 2024 – Present

Bronco STAR

Pomona, CA

- Lead hardware design for a competition rover, designing COTS-implementable drivetrain, power, logic, and communications subsystems, integrating ROS for rover control and distributing tasks across sub-teams.
- Architected a power-distribution system with isolated logic/communications, drivetrain, and arm/science battery domains, using protected high-side switching and digital current/voltage sensing on a locally programmable RP-series microcontroller.
- Designed the Mercury-1 rover control and science-module electronics in KiCad, and developed an embedded control program plus a Linux serial parser for the science module.
- Ported the F´ command-and-data-handling stack (fprime-billee-rcm) to a Raspberry Pi Pico 2 (RP2350) using F´ 4.2.2, Zephyr RTOS 4.3.0, and fprime-zephyr, tuning thread and memory configuration for CCSDS telemetry over USB CDC ACM; verified end-to-end build, flash, and live GDS telemetry.
- As Project Manager (April 2026 – Present), own team budget, deadlines, deliverables, and competition readiness for University Rover Challenge (URC) 2026, managing sub-teams and recruitment.

Technical Skills

Hardware Design: KiCad, Schematic Capture, Multilayer PCB Layout, Cadence OrCAD X, LTspice, ngspice, Gerber/BOM/STEP Release

Embedded Software & Linux: C, C++, Python, F´ Component Architecture, Zephyr RTOS, Yocto/BitBake, Linux Kernel & Device Tree, U-Boot, systemd

Systems Engineering: Integration Testing, Requirements Definition, Design Reviews, Risk Mitigation, Technical Leadership

Tools & Workflow: Git, CMake, Bash, MATLAB, ROS, Simulation & Validation

Languages: English (fluent), Spanish (fluent)

Publications

K. Williams, M. Pham, **L. Lanzillotta**, A. Mariano, J. Pollak, P. Nelson, “An Updated Architecture for a Modular and Scalable Edge Computing System for Small Spacecraft,” *IEEE Xplore*, 2025.